VICTORY 2027

Climate

LECTURE-2 Geography

Sameer Sir



Topics To Be Covered

1. Factors affecting the climate of the area
2. Monsoons and its mechanism
3.
4.

Climate

Distribution of Temperature, Rainfall, winds in Summer and Winter and factors affecting the climate of India (latitudes, altitude, distance from sea, winds, Himalayas, jet streams, and El Nino). Monsoon and its mechanism. Seasons: Summer; Monsoon, Retreating Monsoon and Winter. Temperature and pressure conditions, wind movements, local winds, precipitation and interpretation of climate data.

Diurnal range → Difference between highest and the lowest temperature of a place in single day.

Ex- Bhopal → Max - 42°C
→ min → 27°C

Diurnal range → 15°C

Continentality → A place surrounded by land on all sides.

Maritime influence
Oceanic influence → Influence of water bodies on the temperature of a place.
Moderate temperature.

Rajasthan ← max. 51°C, min. -17°C ✓ (34°C) → Surrounded by land.

Kochi ← max. 32°C, min. 28°C ✓ (4°C) → Located beside Ocean.

Land → Fast heating & Cooling

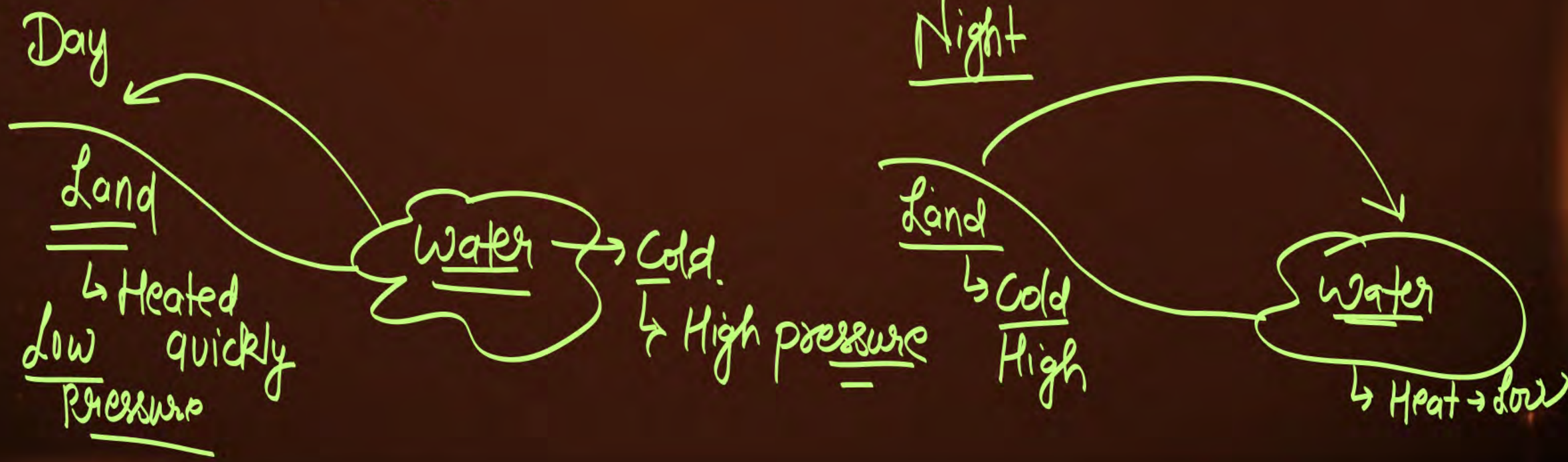
Water → Slow Cooling & heating → Sea Breeze, Land Breeze

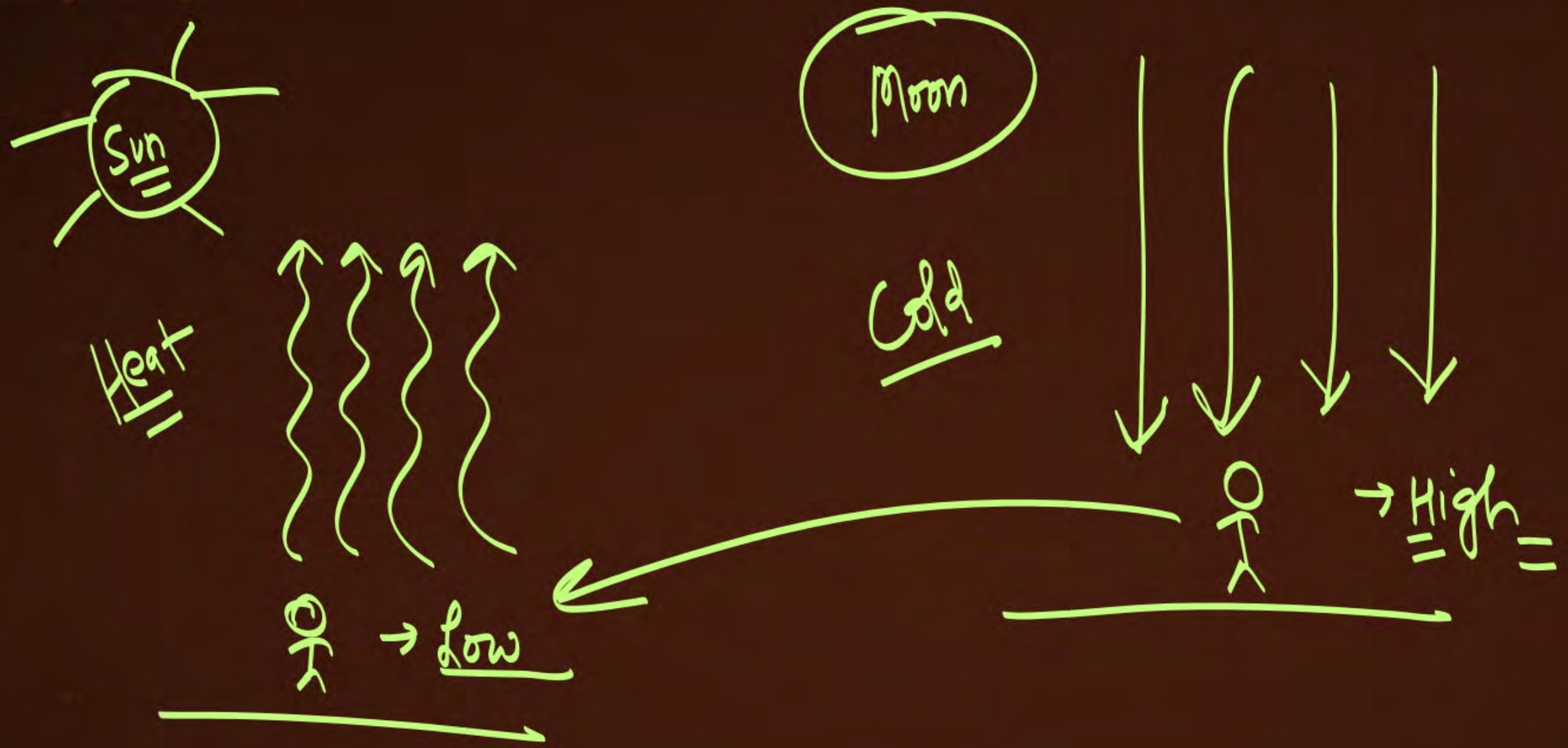
Wind → Pressure difference

High pressure → Low pressure

High pressure → Cold {Low temperature}

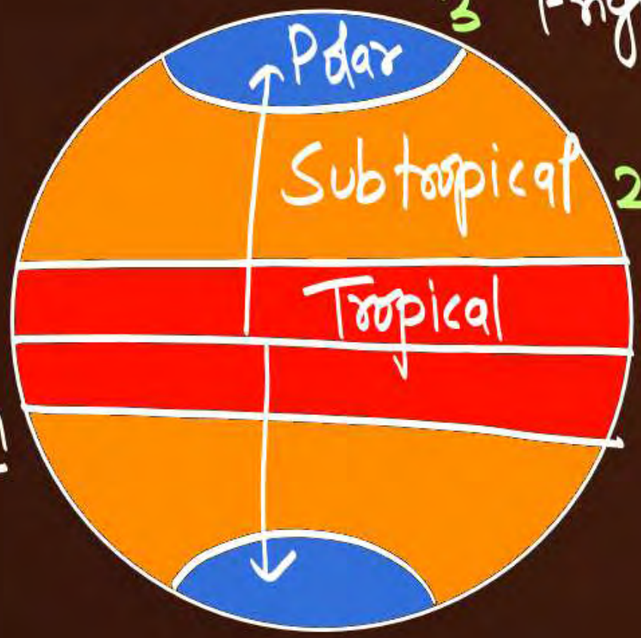
Low pressure → Heat {High temperature}




Sun
Heat
Moon
Cold
→ High
→ Low

India → climate → Tropical monsoon climate

Mausim → Seasonal reversal Winds

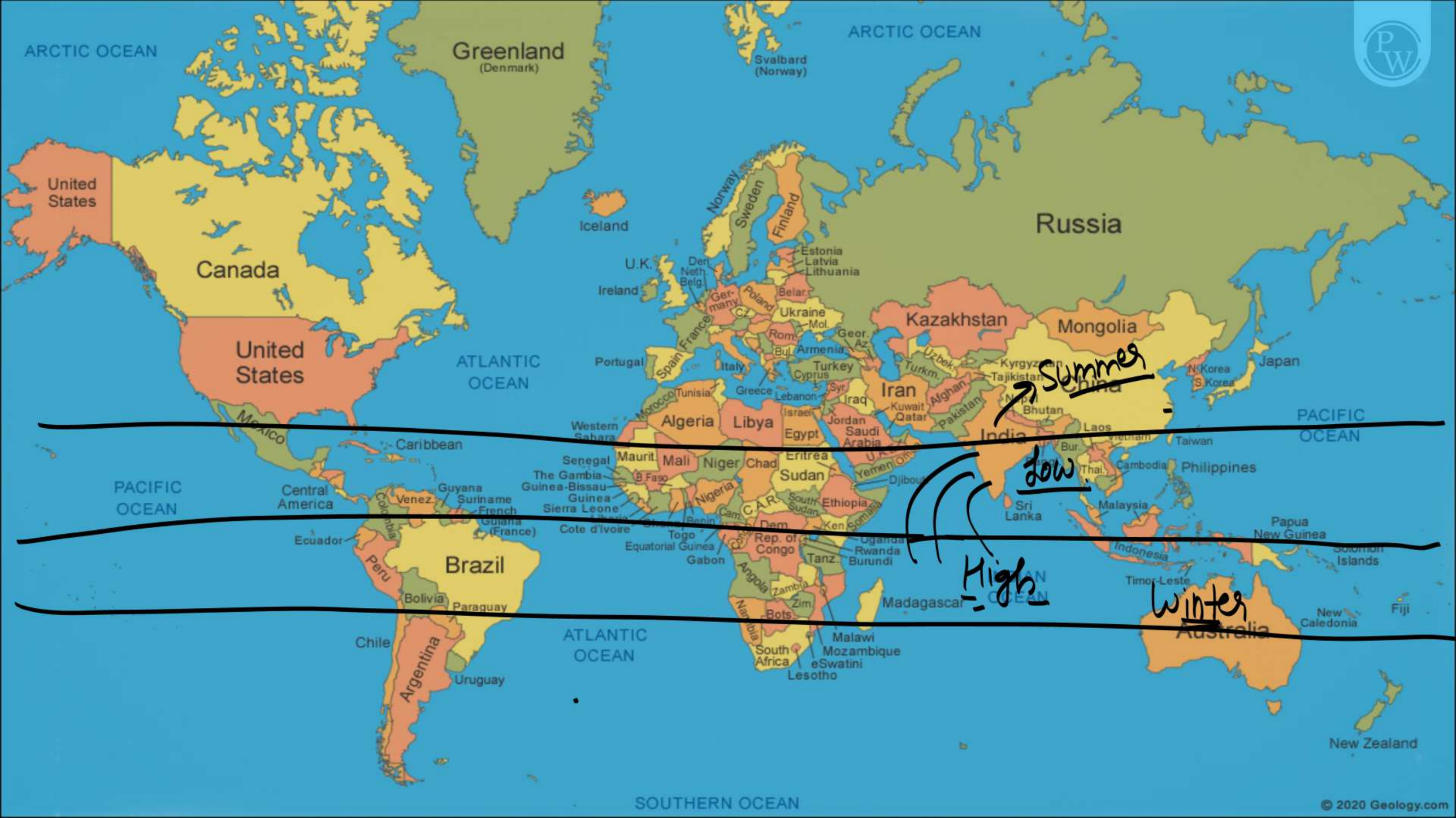


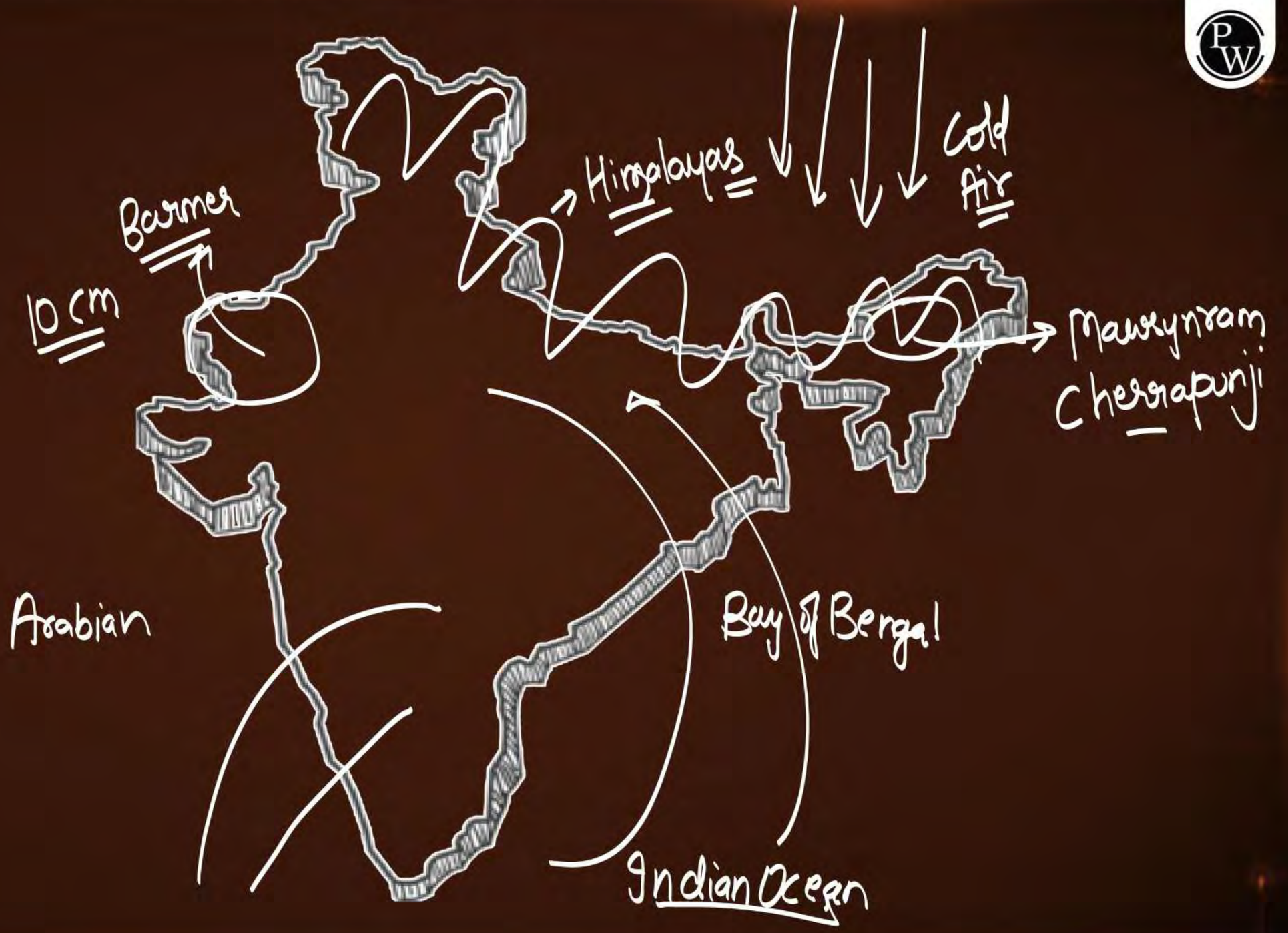
3 Frigid

2 Temperate

1 Torrid

Cancer
Equator
Capricorn


ARCTIC OCEAN
Greenland
(Denmark)
Svalbard
(Norway)
ARCTIC OCEAN
United States
Canada
Iceland
Russia
Norway
Sweden
Finland
Estonia
Latvia
Lithuania
U.K.
Ireland
Den.
Neth.
Belg.
Germany
Poland
Belar.
Ukraine
Mol.
Kazakhstan
Mongolia
United States
ATLANTIC OCEAN
Portugal
Spain
France
Italy
Rom.
Bul.
Armenia
Geor.
Az.
Turkey
Cyprus
Greece
Syr.
Lebanon
Iraq
Iran
Uzbek.
Turkm.
Kyrgyzstan
Tajikistan
Afghan.
Pakistan
Kuwait
Qatar
Summer
China
N. Korea
S. Korea
Japan
Nepal
Bhutan
India
Laos
Vietnam
Taiwan
Bur.
Thai.
Cambodia
Philippines
PACIFIC OCEAN
Morocco
Tunisia
Algeria
Libya
Israel
Jordan
Saudi Arabia
Egypt
Western Sahara
Caribbean
Mexico
Senegal
Maurit.
Mali
Niger
Chad
Eritrea
Sudan
Yemen
Oman
U.A.E.
Djibouti
The Gambia
Guinea-Bissau
Guinea
Sierra Leone
B. Faso
Nigeria
Cam.
C.A.R.
South Sudan
Ethiopia
Somalia
Ken.
Uganda
Rwanda
Burundi
Tanz.
Dem. Rep. of Congo
Congo
Gabon
Equatorial Guinea
Togo
Benin
Cote d'Ivoire
Ghana
Liberia
Low
Sri Lanka
Malaysia
Indonesia
Timor-Leste
Papua New Guinea
Solomon Islands
PACIFIC OCEAN
Central America
Guyana
Suriname
French Guiana (France)
Venez.
Colombia
Ecuador
Peru
Brazil
Bolivia
Paraguay
Chile
Argentina
Uruguay
High
Madagascar
Angola
Zambia
Zim.
Namibia
Bots.
South Africa
Lesotho
eSwatini
Malawi
Mozambique
ATLANTIC OCEAN
Winter
Australia
New Caledonia
Fiji
New Zealand
SOUTHERN OCEAN
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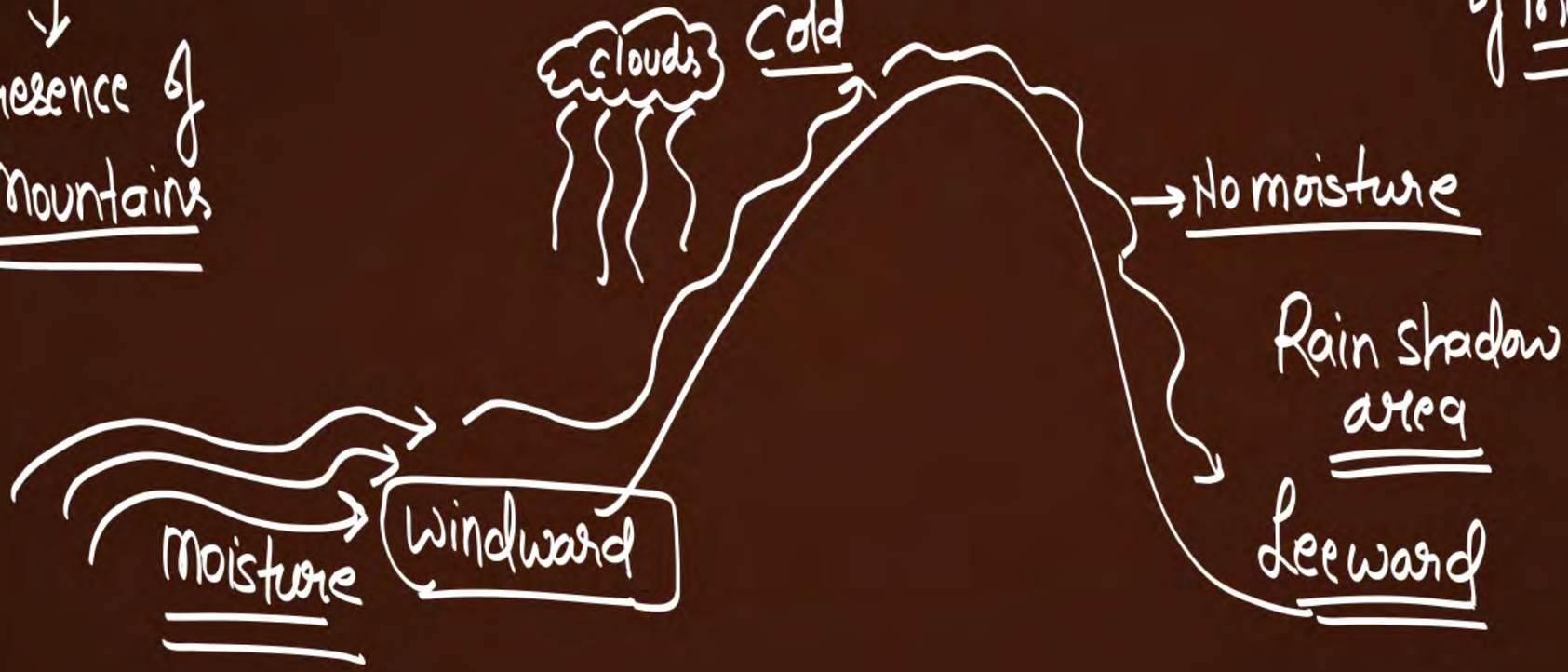

Barmer
10 cm
Himalayas
Cold Air
Mawsynram
Cherrapunji
Arabian
Bay of Bengal
Indian Ocean

Precipitation

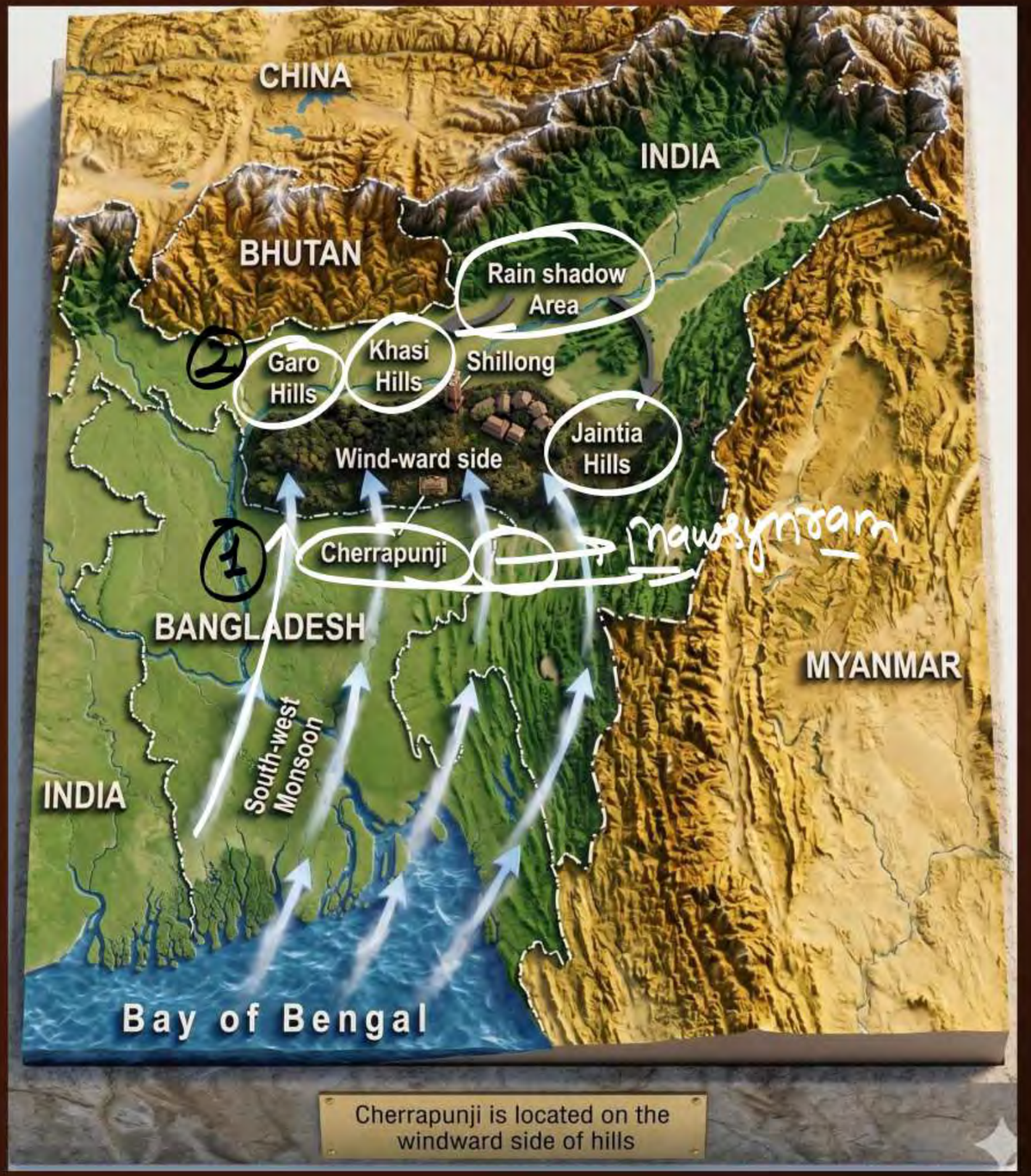
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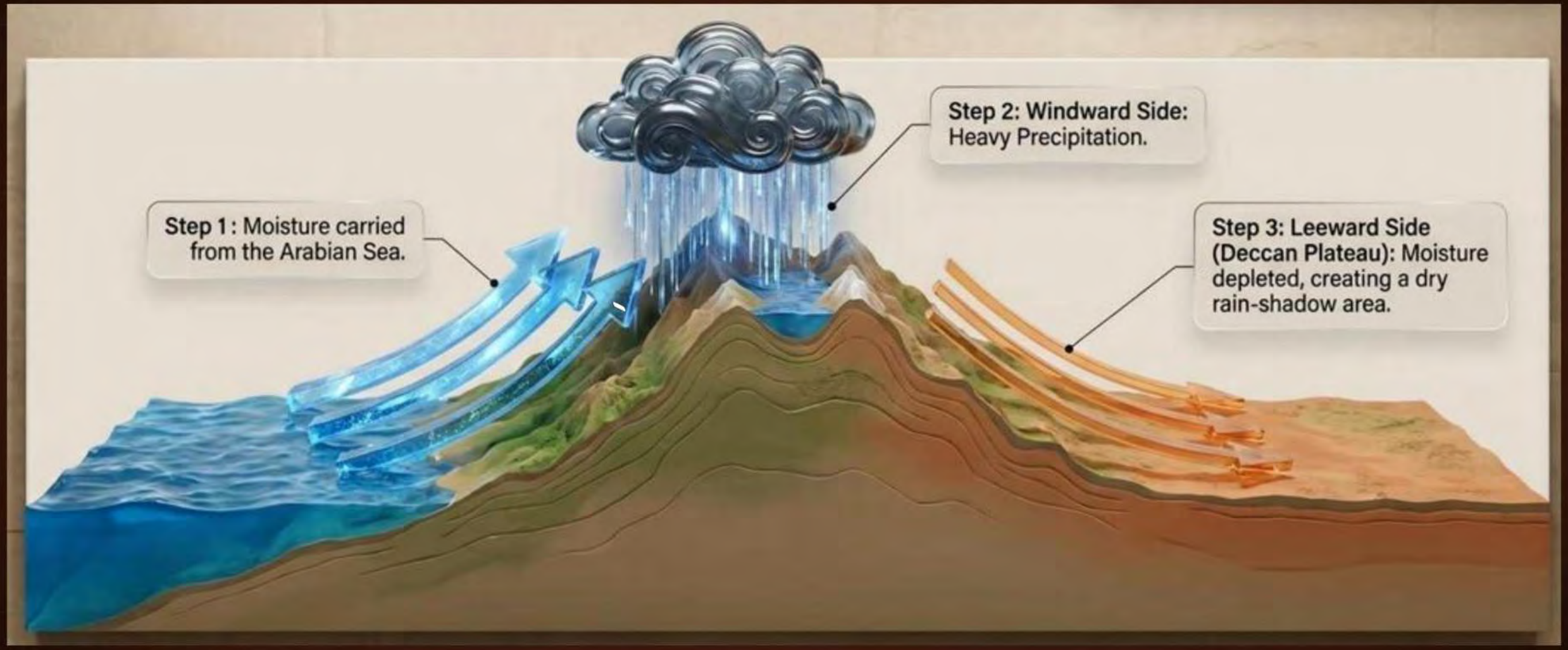
Presence of Mountains

Orographic Rain → Rain caused due to the presence of mountain




INDIA
States and Union Territories
INDIA FLAG
TAJIKISTAN
AFGHANISTAN
PAKISTAN
CHINA (TIBET)
NEPAL
BHUTAN
BANGLADESH
MYANMAR (BURMA)
SRI LANKA
LADAKH
Srinagar
Leh
JAMMU & KASHMIR
Jammu
HIMACHAL PRADESH
Shimla
PUNJAB
CHANDIGARH
Dehradun
UTTARAKHAND
HARYANA
DELHI
New Delhi
Jaipur
RAJASTHAN
UTTAR PRADESH
Lucknow
SIKKIM
Gangtok
Itanagar
ARUNACHAL PRADESH
Dispur
ASSAM
NAGALAND
Kohima
Shillong
MEGHALAYA
Imphal
MANIPUR
Agartala
TRIPURA
Aizawl
MIZORAM
BIHAR
Patna
JHARKHAND
Ranchi
WEST BENGAL
Kolkata
Gandhinagar
GUJARAT
Bhopal
MADHYA PRADESH
CHHATTISGARH
Raipur
ODISHA
Bhubaneswar
BAY OF BENGAL
Daman
DADRA & NAGAR HAVELI AND DAMAN & DIU
MAHARASHTRA
Mumbai
TELANGANA
Hyderabad
Amaravati
Yanam (Puducherry)
ARABIAN SEA
Panaji
GOA
ANDHRA PRADESH
KARNATAKA
Bengaluru
Chennai
Mahe (Puducherry)
PUDUCHERRY
LAKSHADWEEP (INDIA)
Kavaratti
KERALA
TAMIL NADU
Karaikal (Puducherry)
Palk Strait
Thiruvananthapuram
ANDAMAN & NICOBAR ISLANDS (INDIA)
Port Blair
International Boundary
State/UT Boundary
Country Capital
State/UT Capital
Map not to Scale
INDIAN OCEAN
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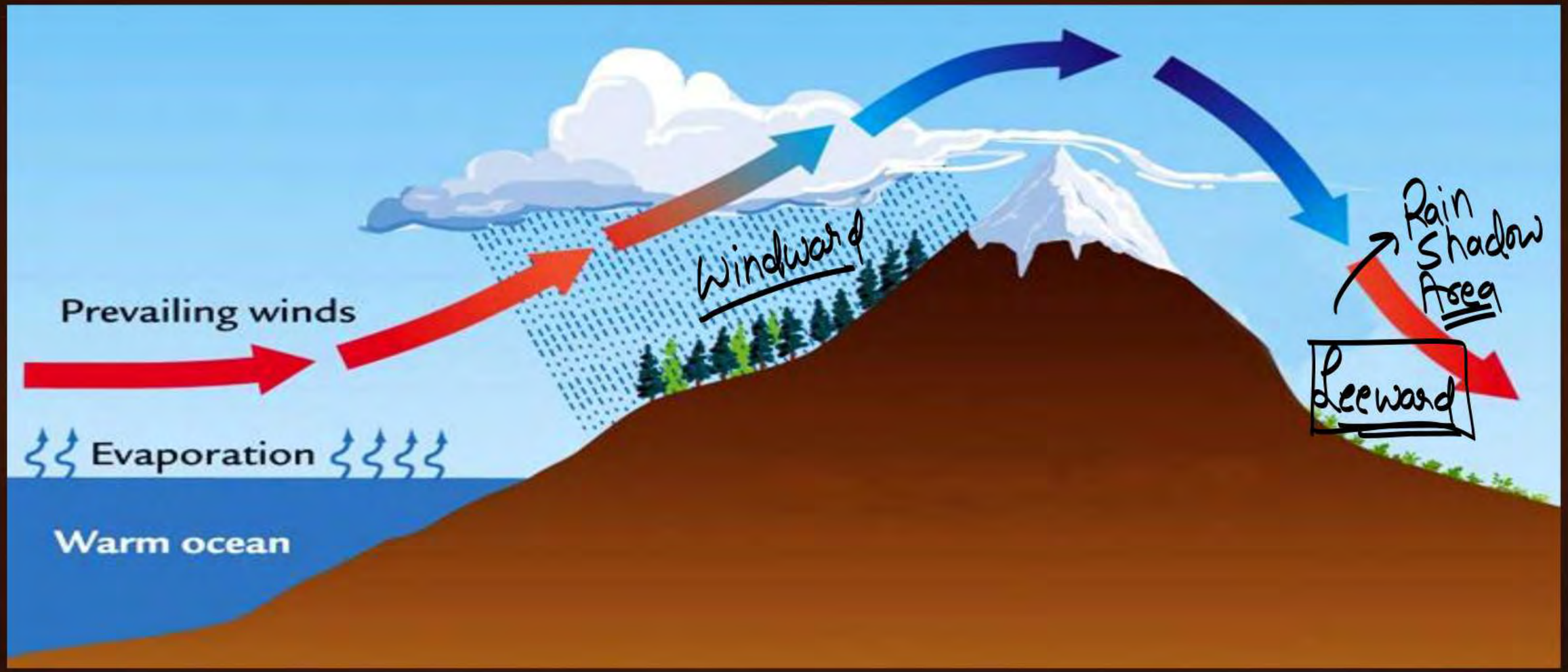

CHINA
INDIA
BHUTAN
Rain shadow Area
Garo Hills
Khasi Hills
Shillong
Jaintia Hills
Wind-ward side
Cherrapunji
Mawsynram
BANGLADESH
MYANMAR
South-west Monsoon
INDIA
Bay of Bengal
Cherrapunji is located on the windward side of hills


Step 1 : Moisture carried from the Arabian Sea.
Step 2: Windward Side: Heavy Precipitation.
Step 3: Leeward Side (Deccan Plateau): Moisture depleted, creating a dry rain-shadow area.

Factors Affecting the Climate of India

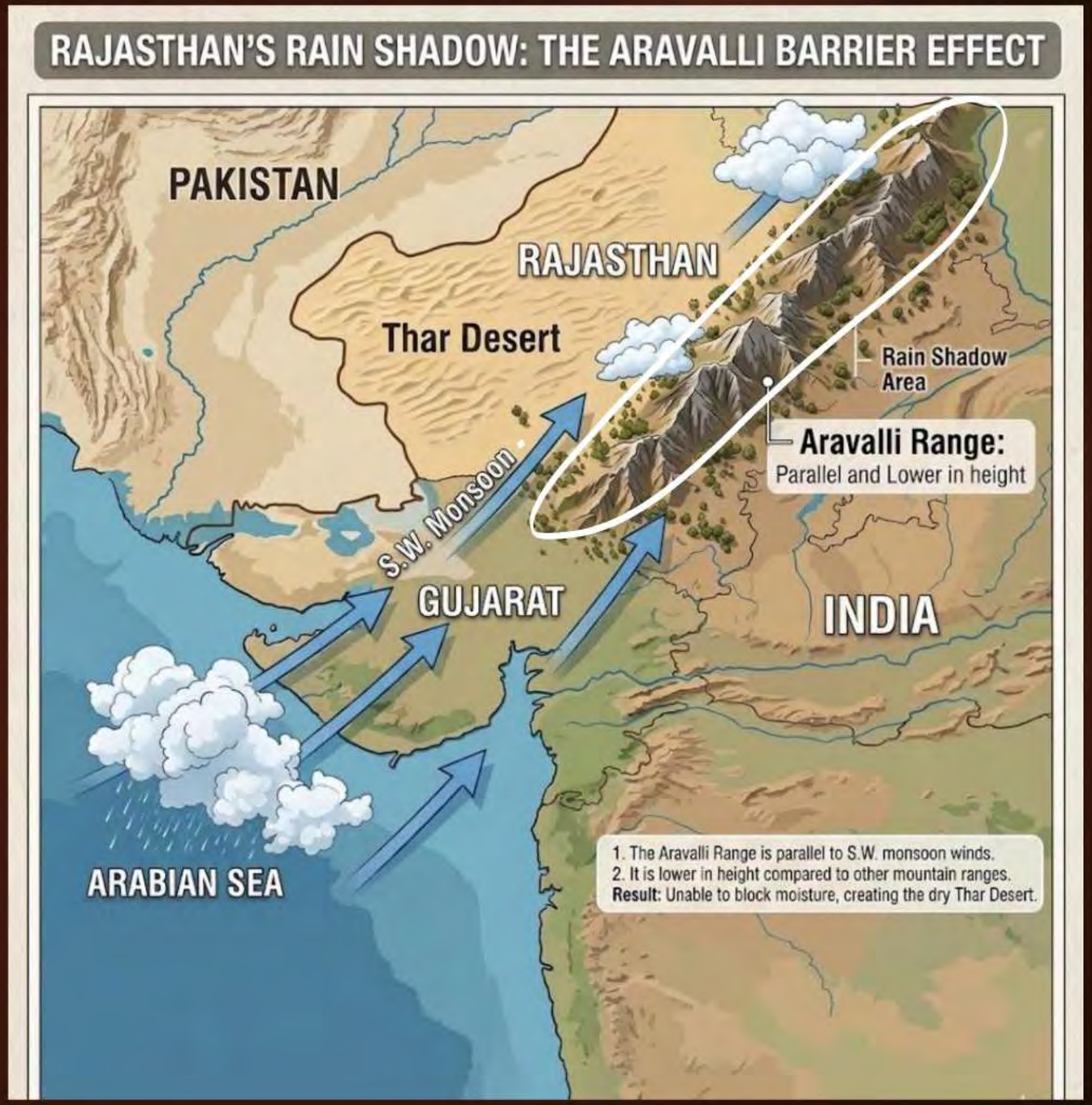
Relief and rainfall: The windward side of Western Ghats receive rainfall from the S-W Monsoon and the windward side of Eastern Ghats receive rainfall from N-E Monsoon.




Prevailing winds
Evaporation
Warm ocean
Windward
Rain Shadow Area
Leeward

No Rainfall in Rajasthan





Himalayas
(The Bouncer)

Latitude
(The Divider)

Altitude
(The Cooler)



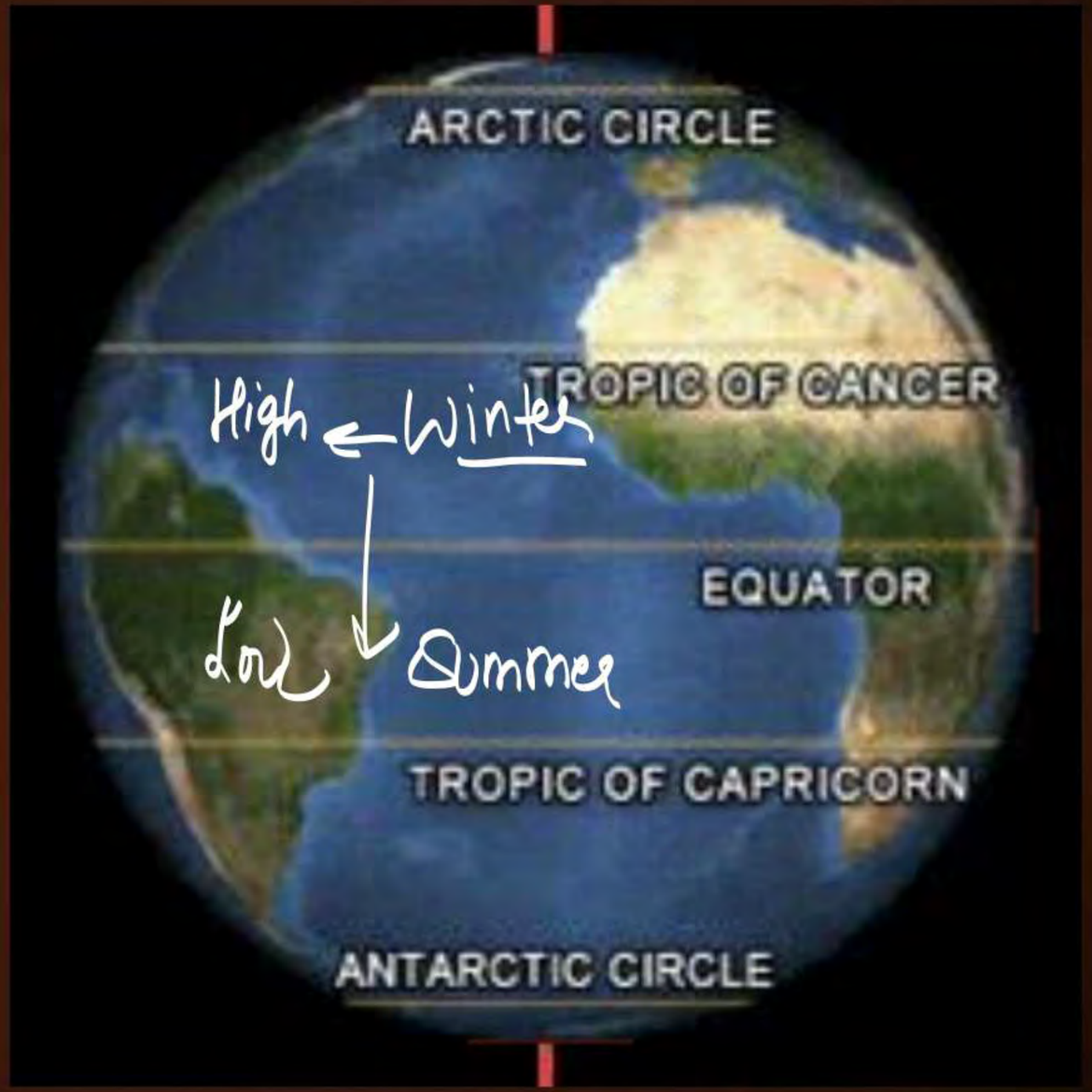
Relief
(The Squeezer)

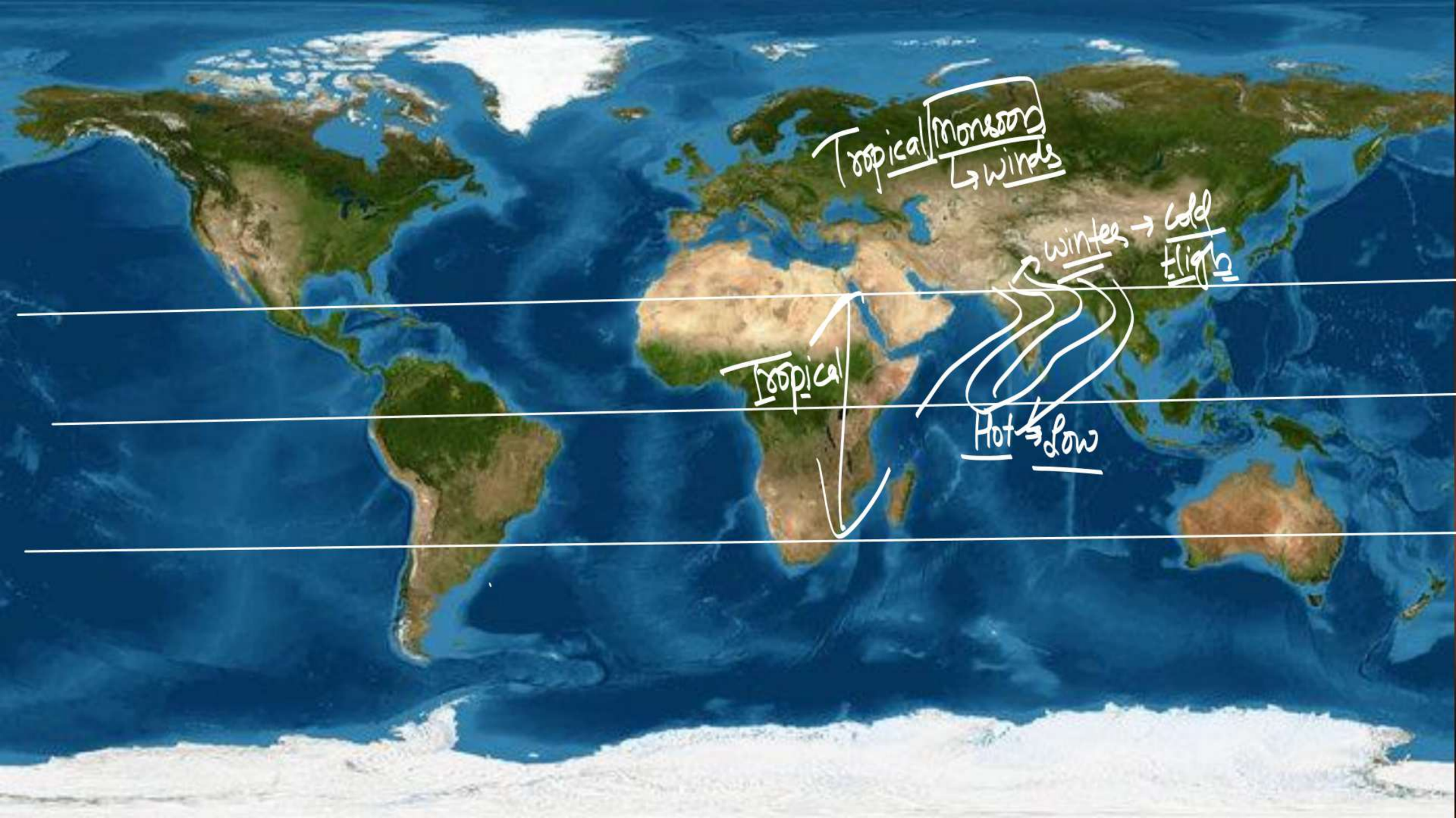
Summer
SW Monsoon
(The Inflow)

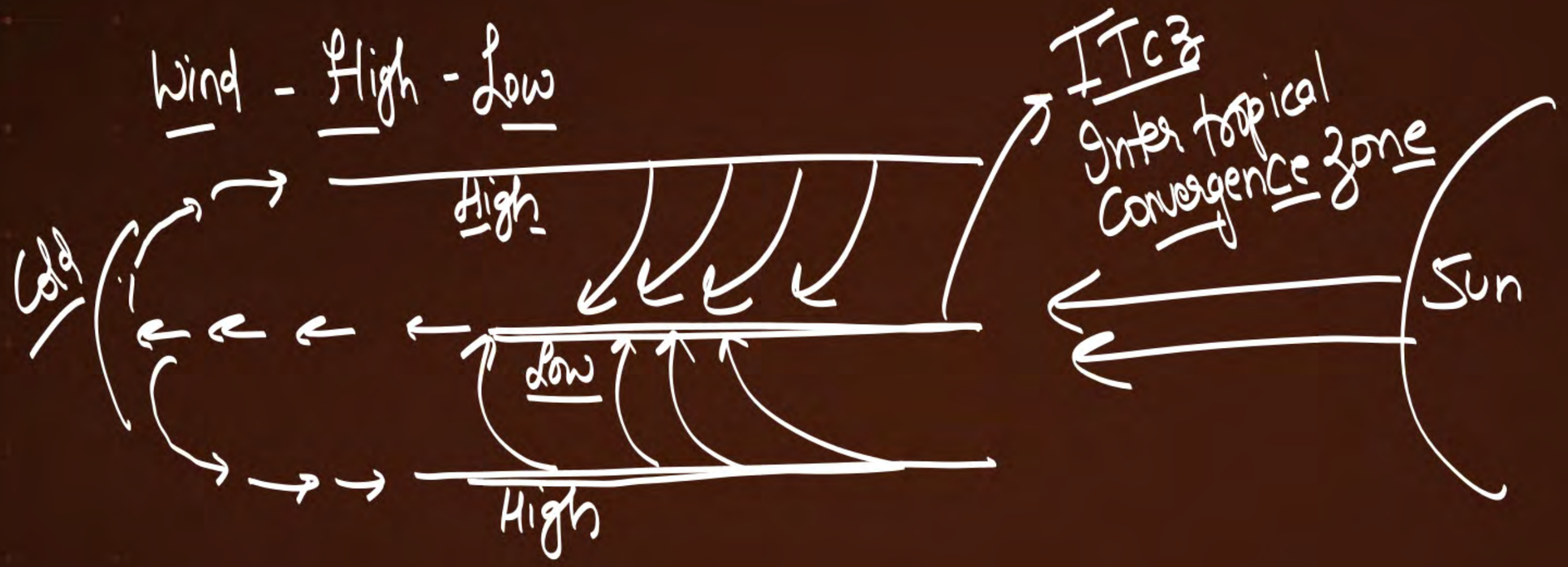
Winter
NE Retreat
(The Outflow)

India doesn't just have one climate; it has an intricate, interconnected ecosystem. Extreme altitudes, mountain shields, and shifting oceanic winds combine every single day to create the most dynamic weather puzzle on Earth.

NotebookLM


ARCTIC CIRCLE
TROPIC OF CANCER
High ← Winter
EQUATOR
Low → Summer
TROPIC OF CAPRICORN
ANTARCTIC CIRCLE

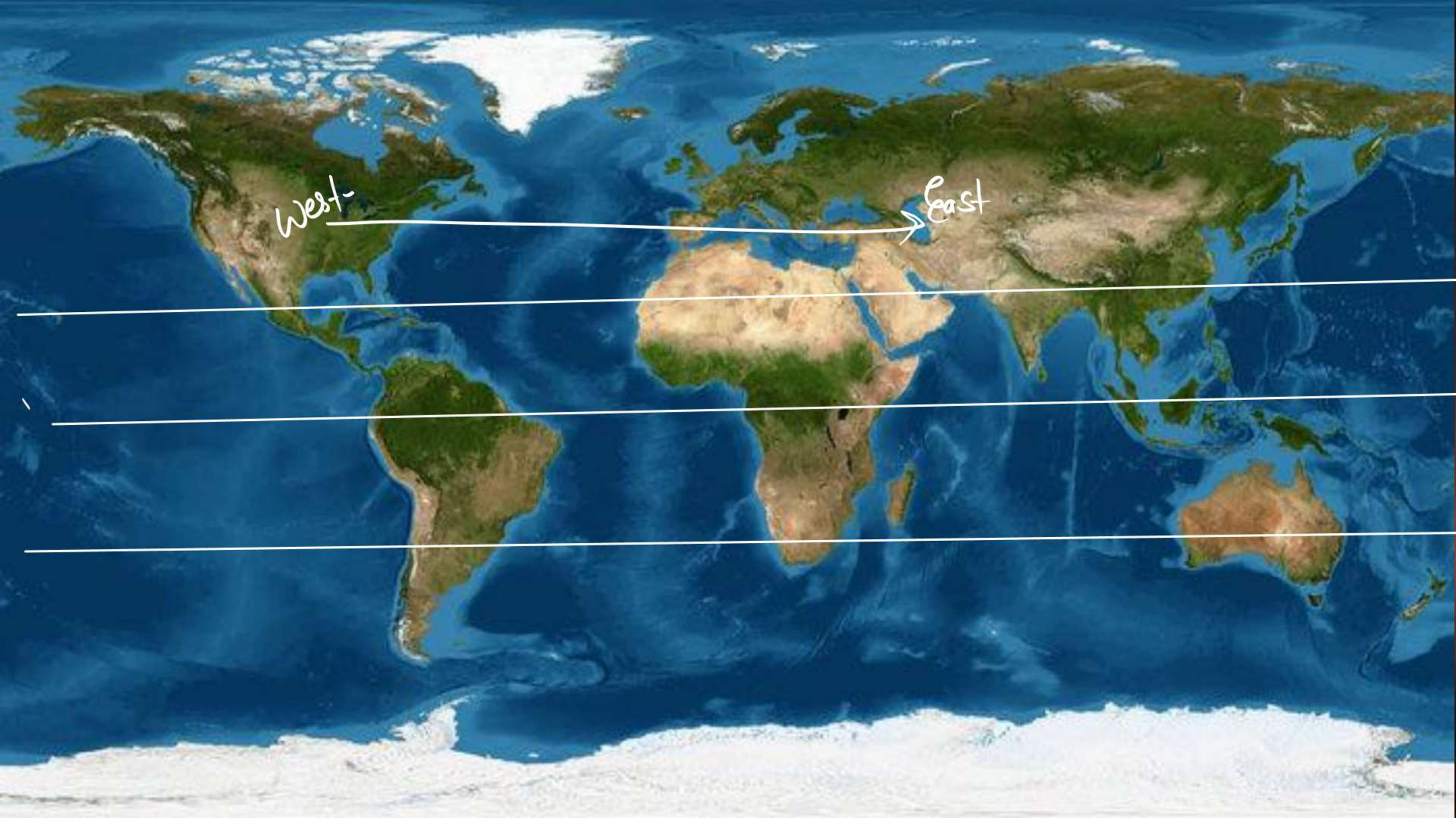

Tropical Monsoon
winds
winter → cold
dry
Tropical
Hot
Low


Wind - High - Low
ITCZ
Inter tropical
Convergence Zone
High
Cold
Low
High
Sun

Factors Affecting the Climate of India

3 **The Monsoon Winds:** The monsoon winds have almost a universal effect in bringing summer rains over the whole of South Asia.

- During summers, winds move quickly into Peninsular India towards the low-pressure system of north west India from the Arabian Sea and Bay of Bengal.
- The moisture carried by these south-westerly winds from the Arabian Sea and Bay of Bengal bring rain to the entire subcontinent, leaving Tibet dry in the rain-shadow' of the Himalayas.

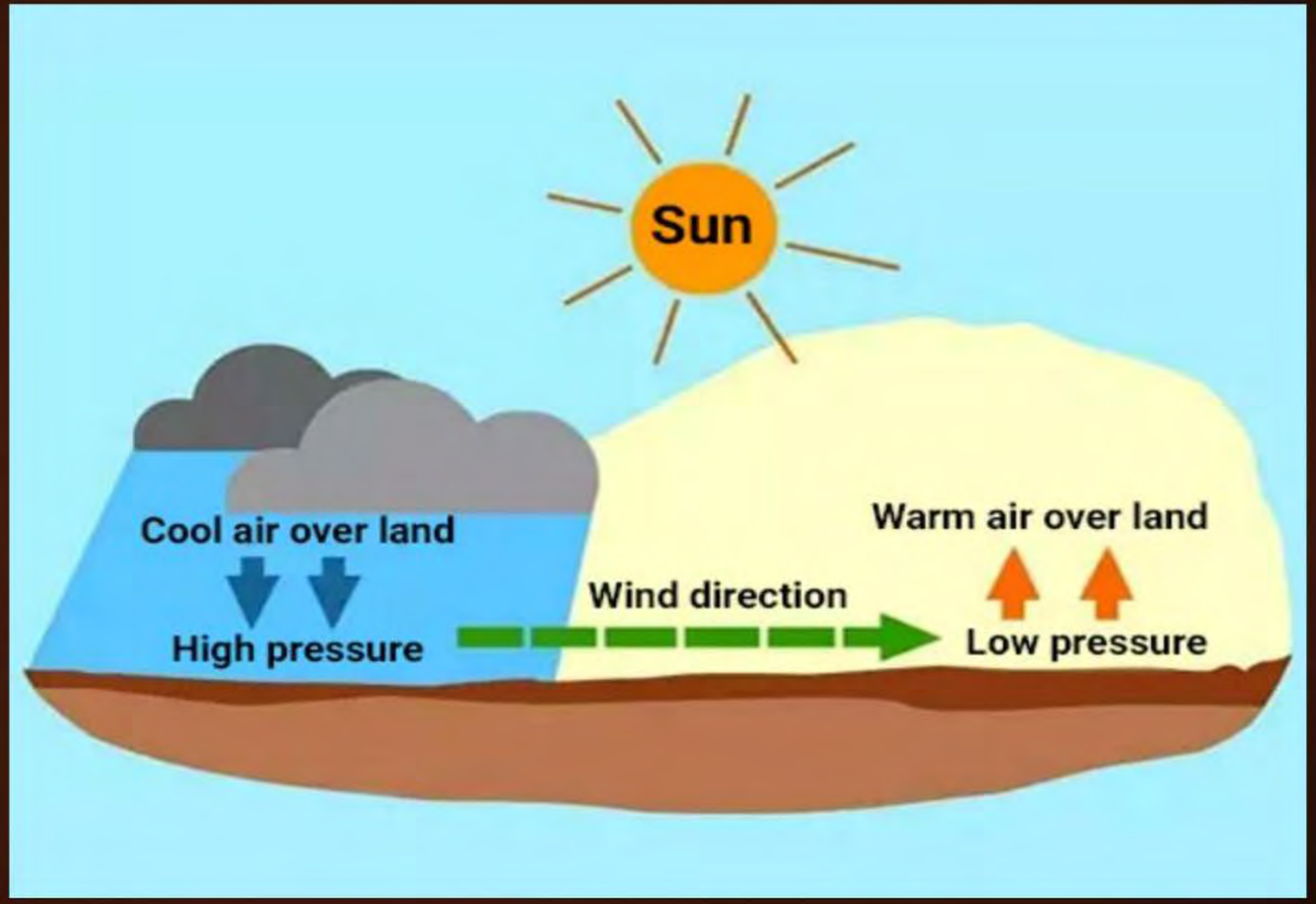

West-
East

Factors Affecting the Climate of India

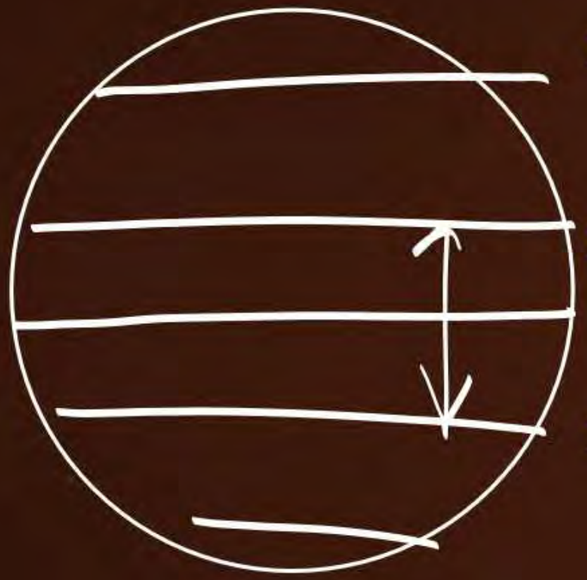
- The south- west monsoon start retreating by the end of September from Punjab and Uttar Pradesh, by October from Central India and by the end of November from South India. Retreating monsoon brings rain in eastern coastal region.

Factors Affecting the Climate of India

- During winter, dry offshore, North-East Monsoon winds blow from high pressure to low pressure region which is centred at the head of Bay of Bengal.
- These winds originate in Mongolia and north west China. As they blow over a vast landmass, they are cold and dry.
- These cold dry winds join the Trade Winds to pass over the Bay of Bengal. They pick moisture from the Bay of Bengal to bring rain to Tamil Nadu in winters.

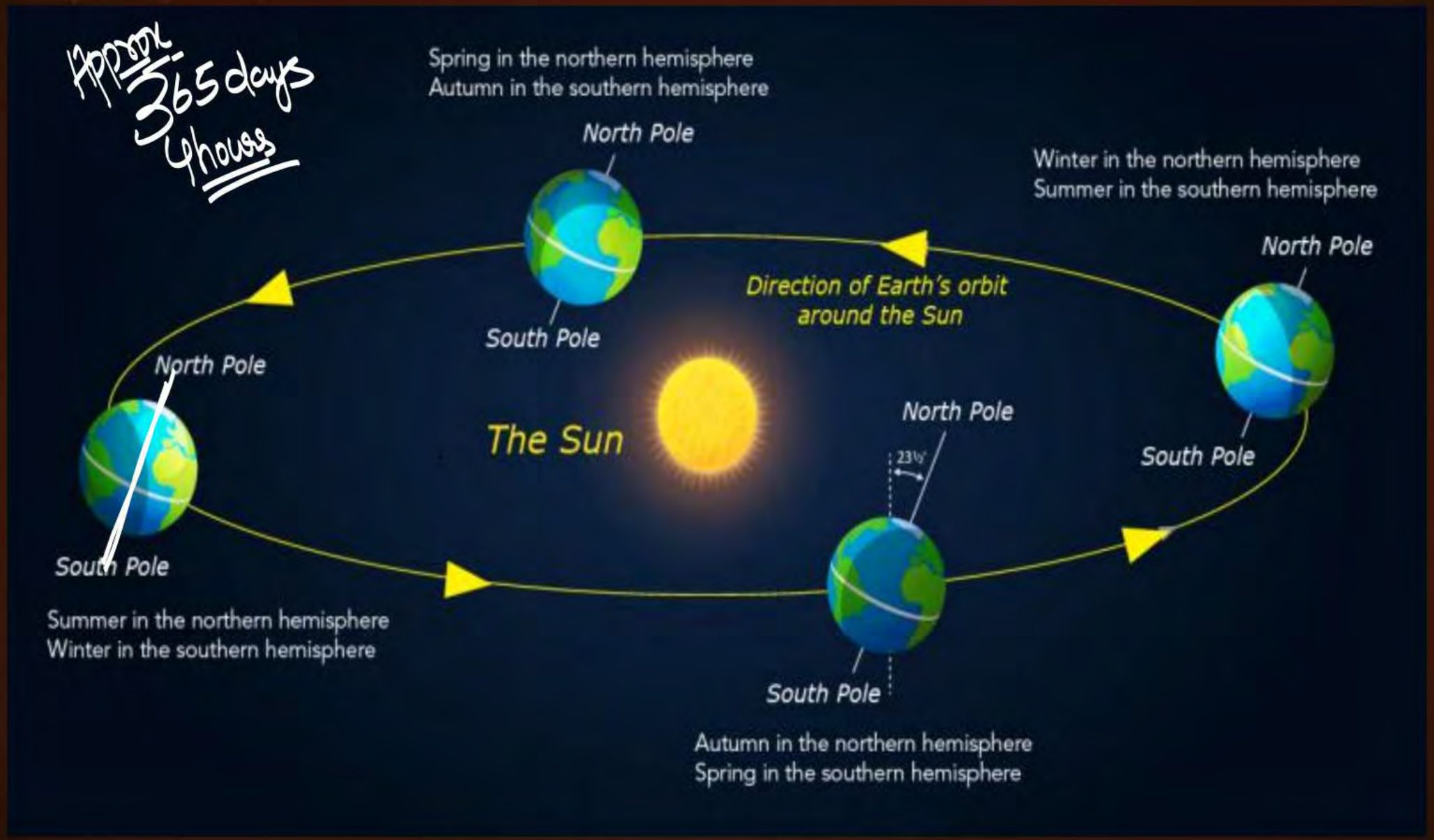

Sun
Cool air over land
High pressure
Wind direction
Warm air over land
Low pressure

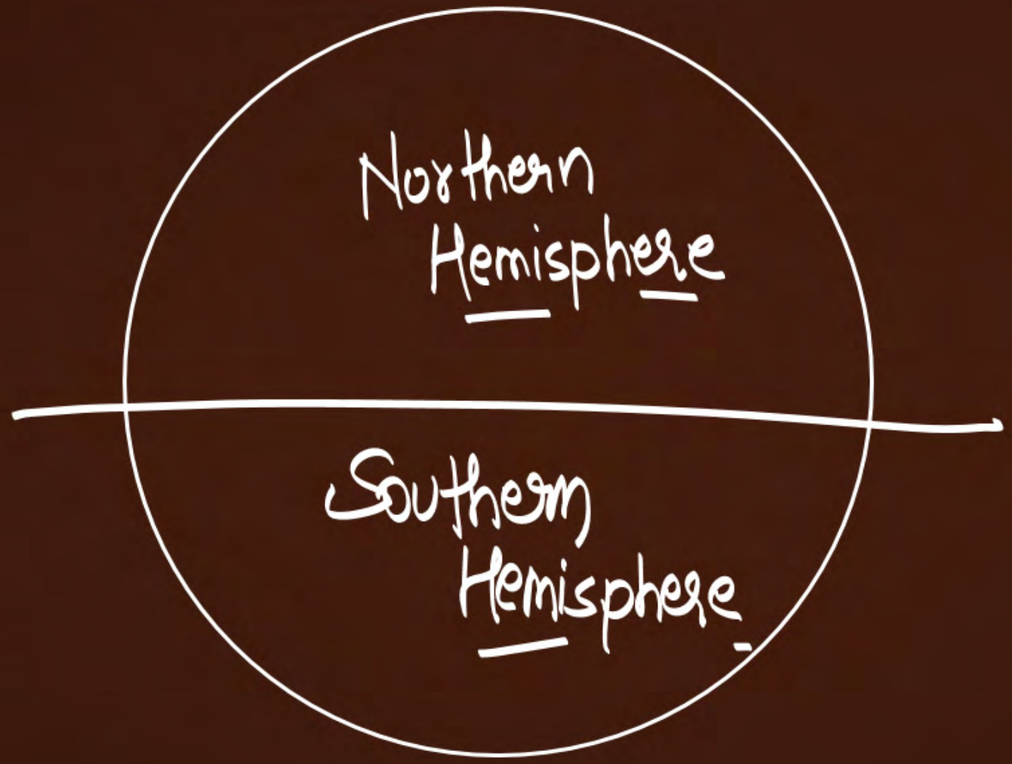
Factors Affecting the Climate of India

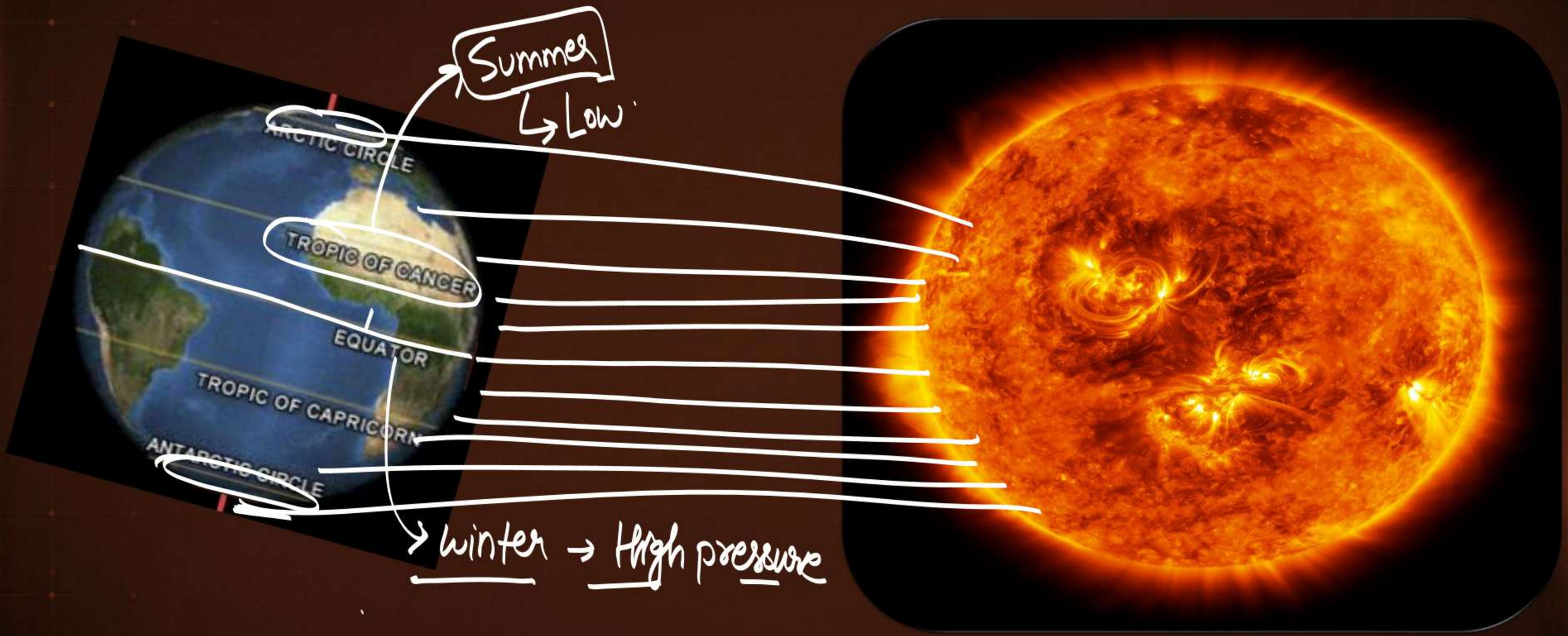


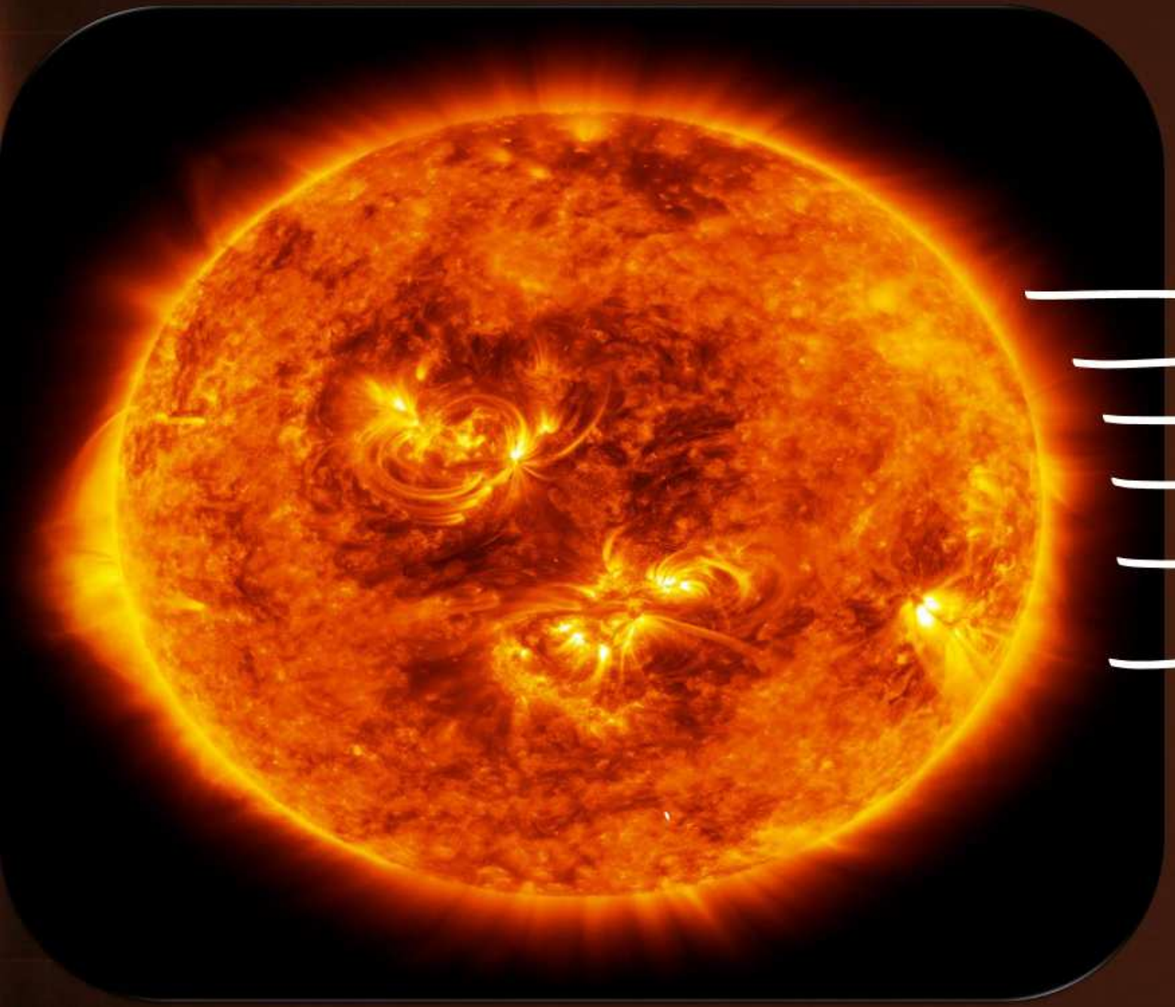
4. **Latitude:** The Indian subcontinent is divided into two parts by the Tropic of Cancer.

- The northern portion lies in the temperate zone and the southern portion lies in the tropical zone.
- The warm temperate or sub-tropical climate of the northern zone gives it cold winter season and hot summer season.
- The southern tropical climatic zone of India is warmer than the north and does not have a clear-cut winter season. Similarly, the northern zone does not have the mid-day sun almost vertically overhead during any part of the year, while the southern zone has the mid-day sun almost vertically overhead at least twice every year.


Approx. 365 days 6 hours
Spring in the northern hemisphere
Autumn in the southern hemisphere
North Pole
South Pole
Winter in the northern hemisphere
Summer in the southern hemisphere
North Pole
South Pole
Direction of Earth's orbit around the Sun
North Pole
South Pole
The Sun
North Pole
23½°
South Pole
Summer in the northern hemisphere
Winter in the southern hemisphere
South Pole
Autumn in the northern hemisphere
Spring in the southern hemisphere

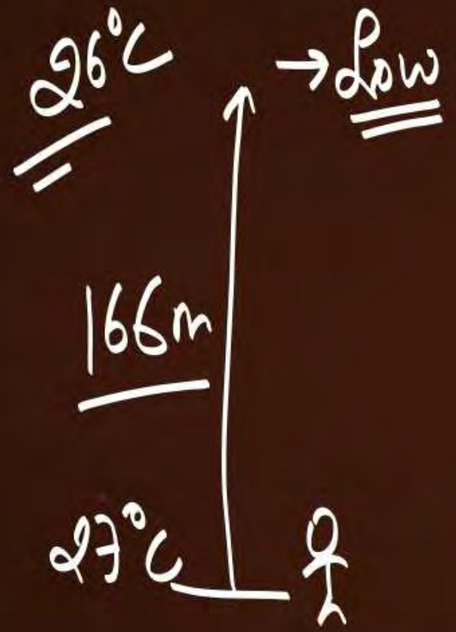

Northern Hemisphere
Southern Hemisphere


Summer
↳ Low
ARCTIC CIRCLE
TROPIC OF CANCER
EQUATOR
TROPIC OF CAPRICORN
ANTARCTIC CIRCLE
winter → High pressure




ARCTIC CIRCLE
TROPIC OF CANCER
EQUATOR
TROPIC OF CAPRICORN
ANTARCTIC CIRCLE
Winter → High
Summer → Low

Factors Affecting the Climate of India



5. **Altitude:** There is a decrease of 1◦ C for every 166 m rise in height, because temperature decreases as you move to higher altitudes.

- So, the mountains are cooler than the plains. For example, Ooty has much lower temperature, than Kochi, because of the altitude.

Factors Affecting the Climate of India

6. **Influence of the Surrounding Seas:** India is flanked by three water bodies, namely, the Indian Ocean, the Bay of Bengal and the Arabian Sea which have a profound impact on the climate, especially in areas lying close to the sea.

- These water bodies act as the major source of moisture to the summer monsoons and bring heavy rainfall to the whole area. In fact, it is because of the nearness to the sea that these areas have a moderate climate.

Maritime influence

Factors Affecting the Climate of India

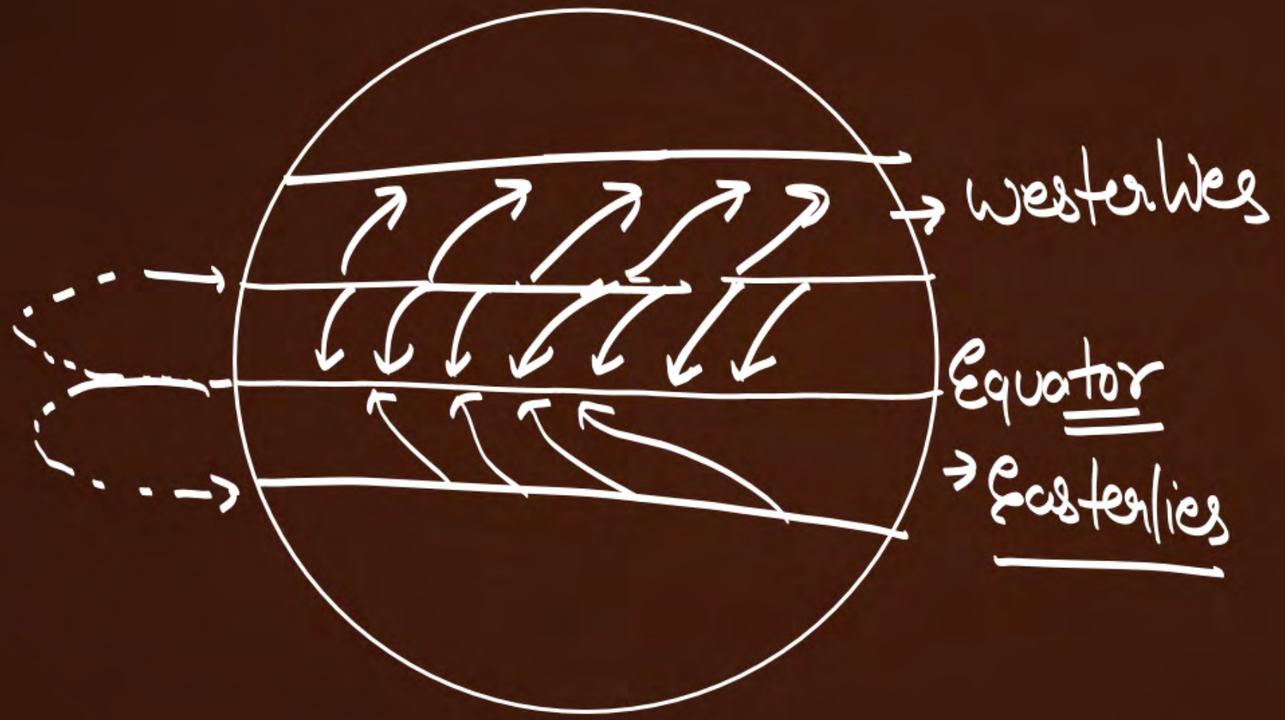
9. **Distance from the Sea:** Areas in the interiors of India have extreme type of climate or continental climate whereas coastal areas have equable or maritime climate. The effect of land breeze and sea breeze caused by differential rate of heating and cooling of land and sea are responsible for moderate climate in coastal areas.
In the Peninsula, the Western Ghats prevent the winds from carrying such benefits far inwards.

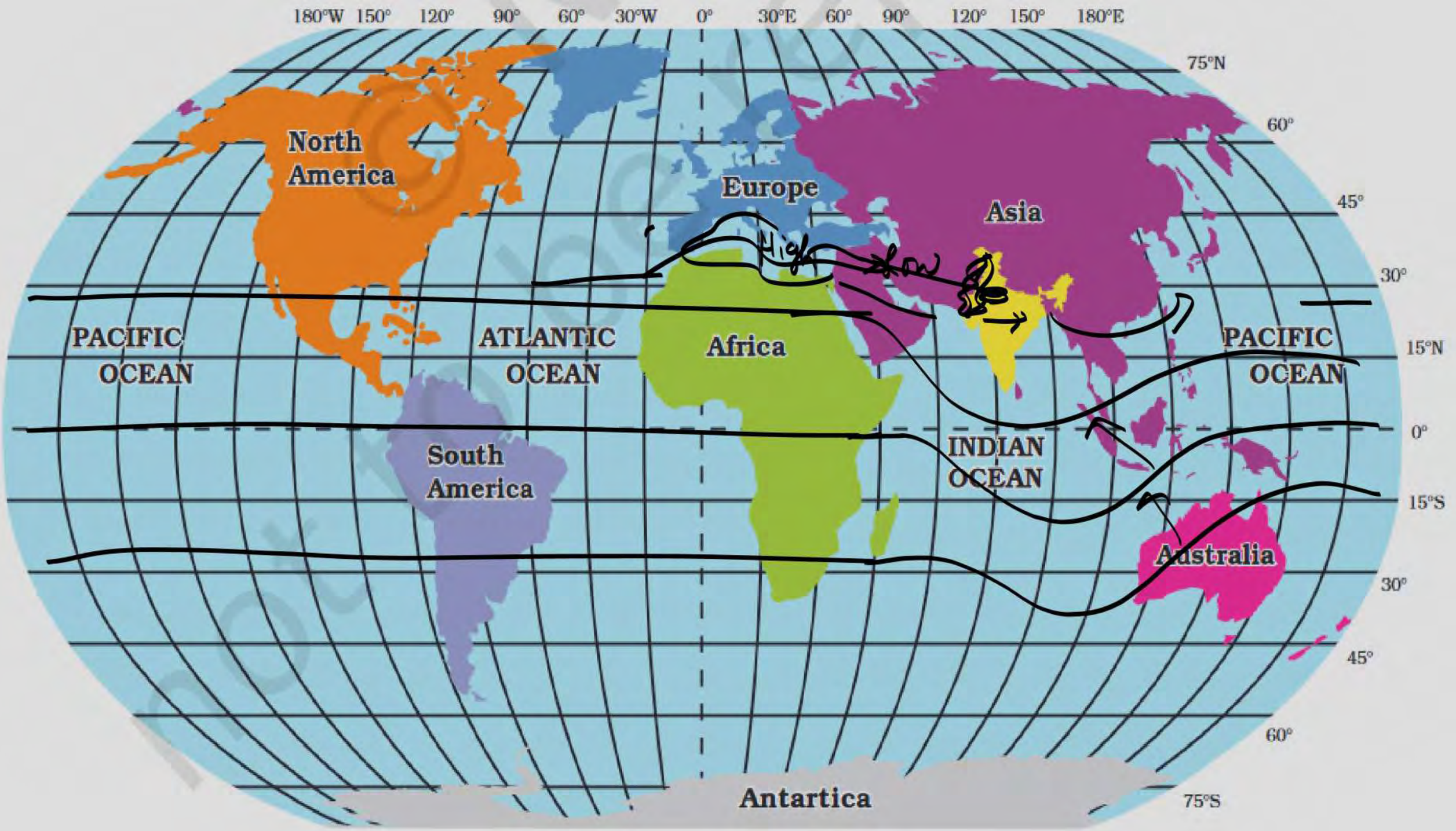
Therefore, the moderating influence of the sea is limited to the coastal areas.

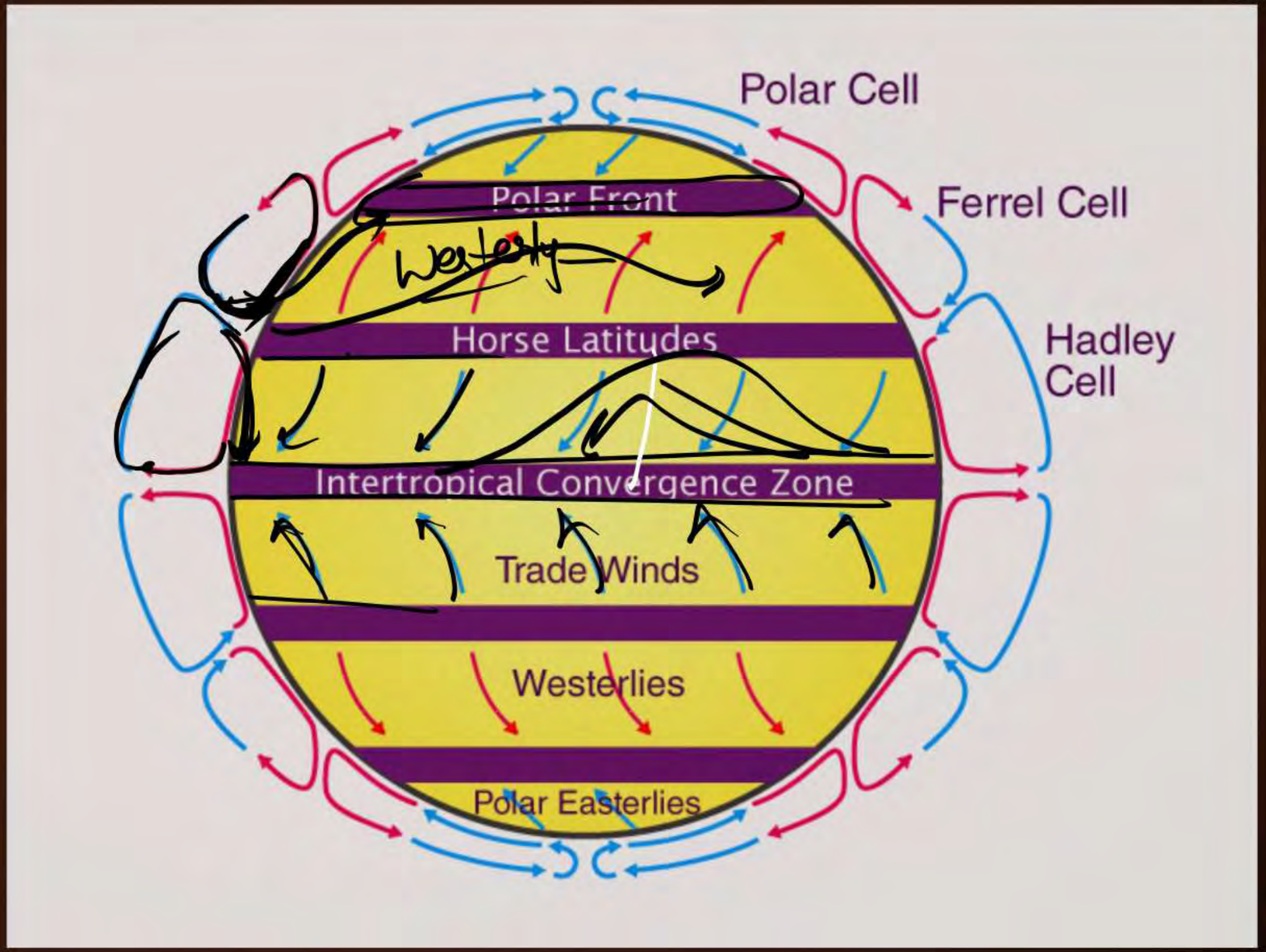
Factors Affecting the Climate of India

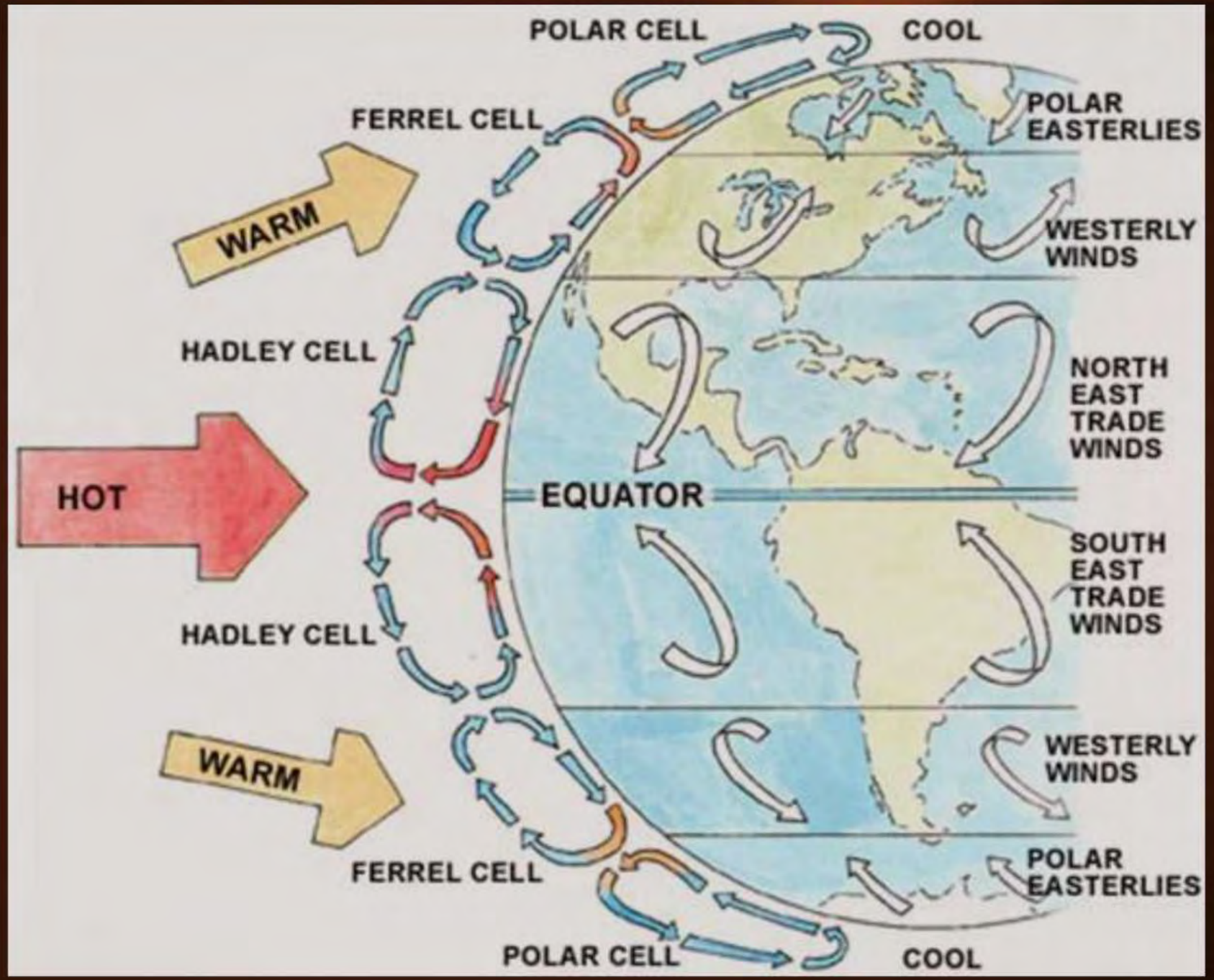
7. **Western Disturbances:** The weather conditions during winter are generally influenced by the distribution pattern of pressure in Central and West Asia.

- In the winter season due to the shifting of the pressure belts, the westerlies blow from 20◦ N to 50◦ N Latitude and north India comes under the influence of the winds and cyclones rising from the Mediterranean Sea. These cyclones bring rain to Northern Plains and snow in Jammu and Kashmir in India.


Westerlies
Equator
Easterlies


180°W 150° 120° 90° 60° 30°W 0° 30°E 60° 90° 120° 150° 180°E
75°N
60°
45°
30°
15°N
0°
15°S
30°
45°
60°
75°S
North America
Europe
Asia
High
Low
PACIFIC OCEAN
ATLANTIC OCEAN
Africa
PACIFIC OCEAN
INDIAN OCEAN
South America
Australia
Antartica


Polar Cell
Polar Front
Ferrel Cell
Westerly
Horse Latitudes
Hadley Cell
Intertropical Convergence Zone
Trade Winds
Westerlies
Polar Easterlies


POLAR CELL
COOL
FERREL CELL
POLAR EASTERLIES
WARM
WESTERLY WINDS
HADLEY CELL
NORTH EAST TRADE WINDS
HOT
EQUATOR
SOUTH EAST TRADE WINDS
HADLEY CELL
WESTERLY WINDS
WARM
POLAR EASTERLIES
FERREL CELL
POLAR CELL
COOL

30% +

30% -

Thank You